BASE CHANGE OF A ONE STEP DEVISSAGE AT A POINT UNDER A RESIDUE-FIELD TRIVIAL POINTED MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the counted span uses no commutative diagram, but the chapter preamble is retained unedited rather than trimmed. No macro was invented or redefined.

Editorial note. Both referenced results live in the same chapter, so their locators point at the \label line inside the pinned chapter file, each recorded with its stable Stacks tag.

Allowed standard prerequisites (editorial).

- One step devissage of a finite type module at a point. Same chapter, Definition environment labelled definition-one-step-devissage-at-x, source line 829 (Stacks tag 05H5). Supplies the definition whose conditions (1)-(4) are checked in the second half of the counted proof; the verification itself stays inside the counted span.
- Base change of a one step devissage over a point. Same chapter, Lemma labelled lemma-base-change-one-step, source line 879 (Stacks tag 05H7). Supplies the over-s' half of the conclusion, invoked at proof lines 942-945; the extra fibre-identification argument for the pointed version is retained in the counted span.

Source reference. Original definition 05H5: sources/stacks-flat/pinned-original-source.tex, source lines 829–829. Same-chapter pointer to the definition invoked in the statement (line 927) and again at proof line 946; it names the four conditions that the counted proof transports, and the transport argument itself is retained verbatim in the delivered body.

Source reference. Original lemma 05H7: sources/stacks-flat/pinned-original-source.tex, source lines 879–879. Same-chapter pointer used twice, in the statement at line 933 and in the proof at line 943, for the over-s' half of the conclusion; the residue-field argument that turns it into the at-a-point statement is the counted obligation and stays in the delivered body.

Lemma 0.1. *Let S, X, $\mathcal{F}$, x, s be as in Definition 05H5. Let $(Z, Y, i, \pi, \mathcal{G}, z, y)$ be a one step dévissage of $\mathcal{F}/X/S$ at x. Let $(S', s') \to (S, s)$ be a morphism of pointed schemes which induces an isomorphism $\kappa(s) = \kappa(s')$. Let $(Z', Y', i', \pi', \mathcal{G}')$ be as constructed in Lemma 05H7 and let $x' \in X'$ (resp. $z' \in Z'$, $y' \in Y'$) be the unique point mapping to both $x \in X$ (resp. $z \in Z$, $y \in Y$) and $s' \in S'$. If S' is affine, then $(Z', Y', i', \pi', \mathcal{G}', z', y')$ is a one step dévissage of $\mathcal{F}'/X'/S'$ at x'.*

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Proof. By Lemma 05H7 $(Z', Y', i', \pi', \mathcal{G}')$ is a one step dévissage of $\mathcal{F}'/X'/S'$ over s'. Properties (1) – (4) of Definition 05H5 hold for $(Z', Y', i', \pi', \mathcal{G}', z', y')$ as the assumption that $\kappa(s) = \kappa(s')$ insures that the fibres $X'_{s'}$, $Z'_{s'}$, and $Y'_{s'}$ are isomorphic to X_s, Z_s, and Y_s. □